BIT300

Integration Technology ALE

EXERCISES AND SOLUTIONS

Course Version: 10
Course Duration: 3 Hours 30 Minutes
Material Number: 50153950

SAP Copyrights, Trademarks and Disclaimers

Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	*Example text*
Window title	*Example text*

Contents

Unit 1
Exercise 1

Set Up ALE Scenarios Using Basic ALE Customizing

Business Example

You want to use ALE to exchange data between two SAP systems. You therefore want to get an overview of the Customizing settings needed to set up ALE scenarios. You also want to check if there is a standard ALE scenario that would be suited to your purposes.

Look at the IMG documentation to see if there are any ALE scenarios for master data distribution between SAP systems. Check the assignment of the logical system names Core, Sales, and Production to the three training clients mentioned by your instructor. Then take a look at the distribution model for the Core logical system.

1. Search in the IMG documentation for ALE scenarios for master data distribution. To do this, use the IMG section *IDoc Interface/Application Link Enabling (ALE)* (transaction `SALE`).

2. In one of the three clients for the SAP ERP logistics and accounting system, check the assignment of the logical system names to the three clients. Make a note of the assignment:

Logical system	Client
Core	
Sales	
Production	

3. In the distribution model for the Core system, select all the model views containing message type MATMAS.

> Hint:
> Use the *Filter model display* button.

4. From the distribution model, display the Core system's partner profiles for the Sales, and Production systems. Check the detail settings for the MATMAS message type.

5. Check the corresponding inbound partner profiles in the Sales system.

Unit 1
Solution 1

Set Up ALE Scenarios Using Basic ALE Customizing

Business Example

You want to use ALE to exchange data between two SAP systems. You therefore want to get an overview of the Customizing settings needed to set up ALE scenarios. You also want to check if there is a standard ALE scenario that would be suited to your purposes.

Look at the IMG documentation to see if there are any ALE scenarios for master data distribution between SAP systems. Check the assignment of the logical system names Core, Sales, and Production to the three training clients mentioned by your instructor. Then take a look at the distribution model for the Core logical system.

1. Search in the IMG documentation for ALE scenarios for master data distribution. To do this, use the IMG section *IDoc Interface/Application Link Enabling (ALE)* (transaction `SALE`).

   a) Select *IDoc Interface/Application Link Enabling (ALE) → Modelling and Implementing Business Processes → Configure Predefined ALE Business Processes → Logistics → Master Data Distribution*.

   b) You will find IMG activities for setting up various master data distribution scenarios.

2. In one of the three clients for the SAP ERP logistics and accounting system, check the assignment of the logical system names to the three clients. Make a note of the assignment:

Logical system	Client
Core	
Sales	
Production	

   a) Choose *IDoc Interface Application Link Enabling (ALE) → Basic Settings → Logical Systems → Assign Logical System to Client*.

   b) One by one, select the table entries for the three training clients and click *Details* for each one. In the *Logical system* field, you will see the logical system name assigned to the client in question (Core, Sales, or Production).

3. In the distribution model for the Core system, select all the model views containing message type MATMAS.

> Hint:
> Use the *Filter model display* button.

a) Select *IDoc Interface/Application Link Enabling (ALE)* → *Modelling and Implementing Business Processes* → *Maintain Distribution Model and Distribute Views*.

b) Click on the *Filter model displays* button, enter **MATMAS** in the *Message Type* field and select *Execute*. The system will only display model views containing message type *MATMAS*.

4. From the distribution model, display the Core system's partner profiles for the Sales, and Production systems. Check the detail settings for the MATMAS message type.

a) In the distribution model's menu, select *Environment* → *Change Partner Profile*.

b) Open the folder for partner type *LS* (logical system) and select the entry for the *SALES* partner.

c) In the outbound parameters, select the entry for message type *MATMAS* and click on *DetailScreenOutb.Parameter*. The *SALES* port and the *MATMAS05* basic type are assigned here. The IDocs should be sent immediately.

5. Check the corresponding inbound partner profiles in the Sales system.

a) Call the distribution model as in step 3 a).

b) In the menu, select *Environment* → *Change Partner Profile*, open the folder for partner type *LS* and select the *CORE* partner.

c) Select the inbound parameters for the entry for message type *MATMAS* and click on *DetailScreenInboundParameter*. The process code *MATM* is assigned here. The IDocs should be processed immediately.

Unit 1
Exercise 2

Display the Documentation for Basic Types

Business Example

You want to use ALE to exchange data between two SAP systems and, by doing so, find out more about the technical implementation of ALE scenarios.

You want to learn about the structure of the *MATMAS05* basic type, so that you can use the Central Material Master Management ALE scenario.

1. From the application menu, display the documentation for the basic type MATMAS05.
2. Find segment type E1MARMM (Material master units of measure) and display the structure. Which segment type is dependent on E1MARMM?
3. Change your user settings so that, in future, the link to the documentation for individual segment fields is displayed next to the link to the structure of the corresponding segment type.

Unit 1
Solution 2

Display the Documentation for Basic Types

Business Example

You want to use ALE to exchange data between two SAP systems and, by doing so, find out more about the technical implementation of ALE scenarios.

You want to learn about the structure of the *MATMAS05* basic type, so that you can use the Central Material Master Management ALE scenario.

1. From the application menu, display the documentation for the basic type MATMAS05.

   a) Select *Tools* → *ALE* → *ALE Administration* → *Services* → *Documentation* → *IDoc Types and Segments*.

   b) In the *Basic type* field, enter **`MATMAS05`** and select *HTML Format*.

2. Find segment type E1MARMM (Material master units of measure) and display the structure. Which segment type is dependent on E1MARMM?

   a) Segment type *E1MARMM* is dependent on segment type *E1MARAM* (*Material master general data*). Click on the *Structure* link below the status information about segment type *E1MARMM* to call a list of segment fields.

   b) Segment type *E1MARMM* (*Master Material European Article*) is dependent on segment type *E1MARMM*.

3. Change your user settings so that, in future, the link to the documentation for individual segment fields is displayed next to the link to the structure of the corresponding segment type.

   a) Choose *Back* to exit the HTML display.

   b) In the transaction's menu, select *Goto* → *User Settings*, click on *Display* <-> *Change*, set the *Documentation Output* indicator and save your changes.

   c) Check the changed setting by displaying the HTML display again. You can now access a *Documentation* link which you can use to call information about the individual segment fields.

Unit 2
Exercise 3

Distribute Master Data

Business Example

In the future, you want all material masters for the entire corporate network to be stored in the central system only and for them to be sent from there to the downstream sales branch and production systems.

> Note:
> In this exercise, when the values include ##, replace the characters with the number that your instructor assigned you.

Create a material master in the Core logical system and send it in IDoc format to the Production and Sales logical systems. Check the inbound and outbound processing for each in the status monitor.

1. Create the *Basic Data 1* and *2* for material MATALE-## with a description of your choice. The material should belong to the *Mechanical Engineering* industry sector and to material type *Finished product*. The base unit of measure is a piece (*PC*).

2. Send your new material using the relevant transaction in the SMD tool to the Production logical system. How many master IDocs and how many communication IDocs were created?

Master IDocs	
Communication IDocs	

3. Send your material again without specifying a target system. How many master IDocs and communication IDocs were created?

Master IDocs	
Communication IDocs	

4. How is the number of IDocs to be sent determined?

____

____

____

5. Display the outbound IDocs for message type MATMAS in the status monitor in the Core logical system. What status do the IDocs have?

> Hint:
> If you want to access one specific IDoc, enter the business object type **`StandardMaterial`** and your material number (**`MATALE-##`**) as the object keys on the status monitor's initial screen.

6. Display the IDocs in the status monitors for each of the two recipient systems, Production and Sales. What status do these IDocs have? Also check the material masters yourself.

Unit 2
Solution 3

Distribute Master Data

Business Example

In the future, you want all material masters for the entire corporate network to be stored in the central system only and for them to be sent from there to the downstream sales branch and production systems.

> Note:
> In this exercise, when the values include ##, replace the characters with the number that your instructor assigned you.

Create a material master in the Core logical system and send it in IDoc format to the Production and Sales logical systems. Check the inbound and outbound processing for each in the status monitor.

1. Create the *Basic Data 1* and *2* for material MATALE-## with a description of your choice. The material should belong to the *Mechanical Engineering* industry sector and to material type *Finished product*. The base unit of measure is a piece (*PC*).
   a) Go to *Logistics* → *Materials Management* → *Material Master* → *Material* → *Create (General)* → *Immediately*.
   b) Enter material number **MATALE-##**, select the industry sector *Mechanical Engineering* and material type *Finished product* and confirm your entries with *Enter*.
   c) Select the views *Basic Data 1* and *Basic Data 2* and click *Enter* again.
   d) Assign a material description of your choice and add the base unit of measure **PC** (*piece*). You do not need to make any entries in the *Basic Data 2* view. Save to create the material master.
2. Send your new material using the relevant transaction in the SMD tool to the Production logical system. How many master IDocs and how many communication IDocs were created?

Master IDocs	**1**
Communication IDocs	**1**

   a) Go to *Tools* → *ALE* → *Master Data Distribution* → *Cross-Application* → *Material* → *Send*.
   b) Enter the following data:

Field Name	Field Value
Material	**MATALE-##**

Field Name	Field Value
Message type	**MATMAS**
Logical system	**PRODUCTION**

Choose *Execute*.

One master IDoc and one communication IDoc have been created.

3. Send your material again without specifying a target system. How many master IDocs and communication IDocs were created?

Master IDocs	**1**
Communication IDocs	**2**

a) Go to *Tools* → *ALE* → *Master Data Distribution* → *Cross-Application* → *Material* → *Send*.

b) Re-enter your material **MATALE-##** and the message type **MATMAS** but leave the *Logical System* field blank. Choose *Execute*.

4. How is the number of IDocs to be sent determined?

The number of communication IDocs depends on the number of receiver systems. If the recipient system is specified in the send transaction, only one communication IDoc is created. If you do not specify any recipient system, then the target systems are determined from the distribution model. This could mean that more than one communication IDoc is created. The number of master IDocs always conforms to the number of master records that are to be sent.

5. Display the outbound IDocs for message type MATMAS in the status monitor in the Core logical system. What status do the IDocs have?

Hint:
If you want to access one specific IDoc, enter the business object type **StandardMaterial** and your material number (**MATALE-##**) as the object keys on the status monitor's initial screen.

a) Go to *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

b) In the *Message Type* field, enter the value **MATMAS** and choose *Execute*.

c) Under *IDoc in outbound processing*, you should see your IDocs with status *03*.

d) Select the entry for message type *MATMAS* and choose *Display IDocs*. You receive a list of all the IDocs for this message type.

e) Select one of your IDocs and choose *Display IDocs*.

6. Display the IDocs in the status monitors for each of the two recipient systems, Production and Sales. What status do these IDocs have? Also check the material masters yourself.

a) Go to *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

b) In the *Message type* field, enter the value **`MATMAS`** and choose *Execute*.

c) Under *IDoc in inbound processing*, you should see your IDocs with status 53.

d) Select the entry for message type *MATMAS* and choose *Display IDocs*. You receive a list of all the IDocs for this message type.

e) Select one of your IDocs and choose *Display IDocs*.

f) Go to *Logistics* → *Materials Management* → *Material Master* → *Material* → *Display* → *Display Current*.

g) Enter your material number (**`MATALE-##`**) and choose *Enter*. Both views are created.

Unit 2
Exercise 4

Create a Reduced Message Type

Business Example

In your enterprise, material masters are created in the head office and sent to production (and elsewhere). The local business department makes changes to specific data views, which you want to protect from being overwritten when the master records are sent again.

Create a reduced message type as a copy of MATMAS and connect it to your distribution model. Then test the configuration by sending a material master.

1. In the Core logical system, create a reduced message type ZMATMAS## with reference to message type MATMAS. In addition to the segments E1MARAM (*Master material general data*) and E1MAKTM (*Master material short texts*), the reduced message type should contain the following segments and segment fields:

Segment	Segment Fields
E1MARAM	*ERNAM* (*Name of Person who Created the Object*)
	BISMT (*Old material number*)
	BRGEW (*Gross Weight*)
	NTGEW (*Net Weight*)
	GEWEI (*Weight Unit*)
E1MTXHM	Select all
E1MTXLM	Select all

> Caution:
> In the segment E1MARAM, ensure the fields LAEDA (*Date of last change*), AENAM (*Name of Person Who Changed Object*), VOLUM (*Volume*) and VOLEH (*Volume unit*) are not selected.

2. You want to use the change pointer function for your reduced message type in the future. Set the appropriate indicator.

3. In the Core logical system's distribution model, create a new model view, TRAINING##, with Core as its sender and Production as its recipient. Use your reduced message type, ZMATMAS##, from the first task.

4. Now have the system generate the outbound partner profiles for message type ZMATMAS## in the Core logical system. What basic type is proposed?

   Basic type: ______________________

5. If the Production logical system is in a different physical system than the Core logical system, then you need to define the reduced message type ZMATMAS## in this system as well before you can distribute the TRAINING## model view.

> Hint:
> You can also transport the reduced message type to the target system.

6. Distribute your TRAINING## model view to the Production partner system. In the Production system, generate the inbound partner profiles for message type ZMATMAS##. Which process code is proposed?

   Process code: ____________

7. Change some values in your material MATALE-## in the Core logical system. Make sure you change the volume and the volume unit.

8. Send the changed material directly to the Production logical system. Display the outbound and inbound IDocs in the status monitor for each system. Did the settings you made in task 1 produce the desired effect?

> Hint:
> Use your reduced message type *ZMATMAS##*.

9. Display the material MATALE-## in the Production system. Which changes have been transferred from the Core system?

10. In the Core system, change the *Old material number* field in your MATALE-## material and then have the system evaluate the change pointers for message type ZMATMAS##. How many communications IDocs are created?

11. Check the result in the Core and Production systems' status monitors. In the segment E1MARAM, what is in the fields BISMT *(Old material number)*, *LAEDA* (*Date of Last Change*), *AENAM* (Name of Person Who Changed Object), and why?

Unit 2
Solution 4

Create a Reduced Message Type

Business Example

In your enterprise, material masters are created in the head office and sent to production (and elsewhere). The local business department makes changes to specific data views, which you want to protect from being overwritten when the master records are sent again.

Create a reduced message type as a copy of MATMAS and connect it to your distribution model. Then test the configuration by sending a material master.

1. In the Core logical system, create a reduced message type ZMATMAS## with reference to message type MATMAS. In addition to the segments E1MARAM (*Master material general data*) and E1MAKTM (*Master material short texts*), the reduced message type should contain the following segments and segment fields:

Segment	Segment Fields
E1MARAM	*ERNAM* (*Name of Person who Created the Object*)
	BISMT (*Old material number*)
	BRGEW (*Gross Weight*)
	NTGEW (*Net Weight*)
	GEWEI (*Weight Unit*)
E1MTXHM	Select all
E1MTXLM	Select all

> Caution:
> In the segment E1MARAM, ensure the fields LAEDA (*Date of last change*), AENAM (*Name of Person Who Changed Object*), VOLUM (*Volume*) and VOLEH (*Volume unit*) are not selected.

a) In the Core logical system, call transaction `SALE` and go to *IDoc Interface/Application Link Enabling (ALE)* → *Modelling and Implementing Business Processes* → *Master Data Distribution* → *Message Reduction* → *Create Reduced Message Type*.

b) In the *Reduced message type* field, enter the value **ZMATMAS##** and choose *Create*.

c) Enter the message type **MATMAS** and select *Enter*. Add a description of your choice and click on *Enter* again.

d) Select the segment *E1MARAM* and then *Choose*.

e) Select *ERNAM*, *BISMT*, *BRGEW*, *NTGEW* and *GEWEI* fields, choose *Select* and then *Enter*.

f) Display the substructure of the *E1MARAM* segment, select the *E1MTXHM* segment (*Master material long text header* and choose *Select*.

g) Display the substructure of the *E1MTXHM* segment, select the *E1MTXLM* subsegment (*Master material long text line)* and choose *Select* again.

h) Save to create the reduced message type. A query appears for a workbench request for transporting the message type to other systems later. Choose *Create Request*, enter a short description of your choice and create the request by saving it.

2. You want to use the change pointer function for your reduced message type in the future. Set the appropriate indicator.

a) To return to the maintenance transaction initial screen, exit the overview of the reduced message type segments.

b) Choose *Activate change pointers*.

3. In the Core logical system's distribution model, create a new model view, TRAINING##, with Core as its sender and Production as its recipient. Use your reduced message type, ZMATMAS##, from the first task.

a) Go to *IDoc Interface/Application Link Enabling (ALE)* → *Modelling and Implementing Business Processes* → *Maintain Distribution Model and Distribute Views* and switch to change mode ().

b) Choose *Create model view*, enter a short text of your choice and the technical name **TRAINING##** and confirm your entries with *Enter*.

c) Select the model view and choose *Add message type*.

d) Enter **CORE** as the sender, **PRODUCTION** as the receiver and **ZMATMAS##** as the message type, and then select *Enter*. Create the new model view by saving it.

4. Now have the system generate the outbound partner profiles for message type ZMATMAS## in the Core logical system. What basic type is proposed?

Basic type: ____________________

a) Select your new model view and in the menu, go to *Environment* → *Generate Partner Profile*.

b) Choose *Execute*. You receive a log containing a message to the effect that outbound partner profiles for message type *ZMATMAS##* and the partner *PRODUCTION* have been created. Exit the log with *Back*.

c) Display one of the two new partner profiles via *Environment* → *Change Partner Profile*. Select the partner *PRODUCTION* (partner type LS).

d) Select the outbound parameters for message type *ZMATMAS##* and click on *DetailScreenOutb.Parameter*. The basic type MATMAS05 is assigned here.

5. If the Production logical system is in a different physical system than the Core logical system, then you need to define the reduced message type ZMATMAS## in this system as well before you can distribute the TRAINING## model view.

> Hint:
> You can also transport the reduced message type to the target system.

a) In the Production logical system, go to *IDoc Interface/Application Link Enabling (ALE) → Modelling and Implementing Business Processes → Master Data Distribution → Message Reduction → Create Reduced Message Type*.

b) Create the reduced message type **ZMATMAS##** as described in step 1.

6. Distribute your TRAINING## model view to the Production partner system. In the Production system, generate the inbound partner profiles for message type ZMATMAS##. Which process code is proposed?

   Process code: ____________

   a) Select your model view and in the menu, go to *Edit → Model view → Distribute*.

   b) Confirm the system selection with *Enter*. You will receive a message informing you that your model view was successfully distributed.

   c) Switch to the Production logical system and call the distribution model from there as well, to check if the copy of your model view is available.

   d) Select the model view, on the menu bar choose *Environment → Generate Partner Profiles* and then *Execute*.

   e) Go back to the distribution model and select *Environment → Change Partner Profile* to check the inbound partner profile for message type *ZMATMAS##*.

   f) Mark the partner Core, in the inbound parameters, select the entry *ZMATMAS##* and choose *DetailScreenInboundParameter*. The process code *MATM* is assigned here.

7. Change some values in your material MATALE-## in the Core logical system. Make sure you change the volume and the volume unit.

   a) Go to *Logistics → Materials Management → Material Master → Material → Change → Immediately*.

   b) Enter **MATALE-##** as your material number choose *Enter*.

   c) Select the *Basic Data 1* view and click on *Enter* again.

   d) Enter a value in each of the fields *Old material number*, *Gross weight* and *Weight unit*, for example. Also add a volume with a suitable unit of measure. Save your changes.

8. Send the changed material directly to the Production logical system. Display the outbound and inbound IDocs in the status monitor for each system. Did the settings you made in task 1 produce the desired effect?

   > Hint:
   > Use your reduced message type *ZMATMAS##*.

   a) Go to *Tools → ALE → Master Data Distribution → Cross-Application → Material → Send*.

b) Enter the material master **MATALE-##**, message type **ZMATMAS##** and the logical system Production and then select *Execute*.

c) Go to *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

d) Enter the message type **ZMATMAS##** and select *Execute*.

e) Select the entry under *ZMATMAS##* and choose *Display IDocs* button.

f) On the next screen, select your IDoc and choose *Display IDocs*.

g) Display the data records and select the segment *E1MARAM*, to check that the system has placed *no-data* characters in the unselected optional fields. You should also be able to see this character in the fields that you did not select in your reduced message type: *VOLUM* (*Volume*) and *VOLEH* (*Volume unit*).

9. Display the material MATALE-## in the Production system. Which changes have been transferred from the Core system?

a) Go to *Logistics* → *Materials Management* → *Material Master* → *Material* → *Display* → *Display Current*.

b) Enter **MATALE-##** as your material number and click on *Enter*.

c) Select the *Basic Data 1* view and click again on *Enter*: the entries in the *Volume* and *Volume unit* fields were not transferred.

d) Check the inbound IDoc in the status monitor as described in steps 8. c) to g).

10. In the Core system, change the *Old material number* field in your MATALE-## material and then have the system evaluate the change pointers for message type ZMATMAS##. How many communications IDocs are created?

a) Call the material master as described in step 7 to change it.

b) Enter a value in the *Old material number* field and save the change.

c) Go to *Tools* → *ALE* → *ALE Administration* → *Services* → *Change Pointers* → *Evaluate*.

d) Enter your message type **ZMATMAS##** and choose *Execute*.

e) IDocs are generated for all materials that were changed since the change pointer function was activated for your message type.

11. Check the result in the Core and Production systems' status monitors. In the segment E1MARAM, what is in the fields BISMT *(Old material number)*, *LAEDA* (*Date of Last Change*), *AENAM* (Name of Person Who Changed Object), and why?

a) Call one of the outbound IDocs for message type ZMATMAS## in the status monitor in the Core system. Proceed as in step 8.

b) The change you have made is visible in the field *Old material number*, because you selected this field in your reduced message type. The fields *Date of Last Change* and *Name of Person Who Changed Object* contain /. This is because you did not select these fields when creating the reduced message type.

c) To finish, display an inbound IDoc for message type *ZMATMAS##* in the Production system. The *Old material number* field contains the changes you made, the *Date of last change* and *Last changed by* fields contain /.

Unit 3
Exercise 5

Create Partner Profiles and Condition Record

Business Example

In the future, you want to send purchase orders to one of your vendors by means of EDI.

Create partner profiles for the vendor and then a suitable determination record for the message type of the purchase order data output.

1. Create partner profiles with the vendor T-K515D## for the message type ORDERS. Use the IDoc basic type ORDERS05 and the file port SUBSYSTEM. The IDocs should be transferred to the subsystem together, but this should not be started by the SAP system. The appropriate message type has the key NEU. You work on the application level of Purchasing. Which process code is specified? Assign yourself as the agent in the case of an error.

   Process code: ______________

   > Hint:
   > Please also remember the partner type and the partner function.

2. Get the system to check the partner profiles you have just created.

3. In the application menu of Purchasing (purchase orders), create a determination record for the message type NEU. Messages of this message type should always be output if the purchasing organization 1000 orders goods from the vendor T-K515D##. The system should output the message in IDoc format by means of EDI. The IDoc should be generated immediately after a purchase order is created.

4. Optional: How do you proceed if you also have an agreement with your vendor T-K515D## that in future he should send you order confirmations by EDI for your purchase orders?

   > Hint:
   > If necessary, use the input help for the *Message type* field to look for a suitable message type using the short description Order confirmation. You can also use the short description to determine the appropriate process code.

Create Partner Profiles and Condition Record

Business Example

In the future, you want to send purchase orders to one of your vendors by means of EDI.

Create partner profiles for the vendor and then a suitable determination record for the message type of the purchase order data output.

1. Create partner profiles with the vendor T-K515D## for the message type ORDERS. Use the IDoc basic type ORDERS05 and the file port SUBSYSTEM. The IDocs should be transferred to the subsystem together, but this should not be started by the SAP system. The appropriate message type has the key NEU. You work on the application level of Purchasing. Which process code is specified? Assign yourself as the agent in the case of an error.

   Process code: _______________

   > Hint:
   > Please also remember the partner type and the partner function.

   a) Choose *Tools* → *ALE* → *ALE Administration* → *Runtime Settings* → *Partner Profiles*.

   b) Choose *Create*, enter the partner number **`T-K515D##`** and the partner type **`LI`** and save your entries.

   c) In the output parameter area, choose *Create outbound parameter*.

   d) Add the partner role **`LF`** and the message type **`ORDERS`**. On the *Outbound Options* tab page, add the receiver port **`SUBSYSTEM`** and the IDoc basic type **`ORDERS05`**. If they are not already selected, choose *Collect IDocs* and *Do not start subsystem*.

   e) Choose the *Message Control* tab page and choose *Insert row*.

   f) Enter the application **`EF`** and the message type **`NEU`**. In the input help for the field *Process code*, ME10 (*ORDERS: Purchase order*) appears. Press *Enter* to accept this key.

   g) On the *Post Processing: Permitted Agent* tab page, enter agent type US in the Ty. field. Add your user BIT300-## and a notification language of your choice. Save these changes.

2. Get the system to check the partner profiles you have just created.

   a) If necessary, open the folder for the partner type LI and select your vendor T-K515D##.

   b) Choose *Check*, then choose *Execute* in the window displayed: you obtain a detailed log of the check on your settings.

3. In the application menu of Purchasing (purchase orders), create a determination record for the message type NEU. Messages of this message type should always be output if the purchasing organization 1000 orders goods from the vendor T-K515D##. The system should output the message in IDoc format by means of EDI. The IDoc should be generated immediately after a purchase order is created.

   a) Choose *Logistics* → *Materials Management* → *Purchasing* → *Master Data* → *Messages* → *Purchase Order* → *Create*.

   b) Enter the output type **NEU** and choose *Enter*.

   c) Choose the key combination *Purchasing Output Determination: Purch. Org./Vendor for EDI* and confirm your choice with *Enter*.

   d) Enter the purchasing organization **1000**, the vendor **T-K515D##**, the partner function **VN**, the medium **6** and the dispatch time **4**. Create the determination record by saving.

   > Hint:
   > If you leave the *Language* field empty, the system uses the language of the vendor in his master record.

4. Optional: How do you proceed if you also have an agreement with your vendor T-K515D## that in future he should send you order confirmations by EDI for your purchase orders?

   > Hint:
   > If necessary, use the input help for the *Message type* field to look for a suitable message type using the short description Order confirmation. You can also use the short description to determine the appropriate process code.

   a) Choose *Tools* → *ALE* → *ALE Administration* → *Runtime Settings* → *Partner Profiles*.

   b) Open the folder for the partner type LI and select your vendor T-K515D##.

   c) In the inbound parameter area, select *Create inbound parameters* and first enter the partner role **LF** again.

   d) Call up the input help for the field *Message type* and open the details in the *Restrictions* tab page by clicking on the small triangle in the bar below the tab page.

   e) In the field *Short text*, enter ***Order confirmation*** and press *Enter* to confirm: The system prompts you with, among other things, the message type *ORDRSP* (*Purchase order/order confirmation*). Select the default and accept it with *Enter*.

   f) Now call up the input help for the field *Process code* and follow the short description of the process: The suitable process code has the key *ORDR* (*ORDRSP Purchase order confirmation*). Accept it with a double-click and save your changes.

Unit 3
Exercise 6

Send a Purchase Order to a Vendor

Business Example

The central purchasing department of the company orders goods from a vendor by means of an EDI connection.

Create a purchase order to an external vendor in the system at company headquarters and check whether an IDoc was generated with the data of this purchase order.

1. In the CORE system, create a purchase order at the vendor T-K515D## for 100 pcs. of the material T-M15A## for the plant 1000. You work for the purchasing organization 1000 and the purchasing group 001 in the company code 1000. Which message type is found, and how is the message to be processed?

   Message type: _______

   Transmission medium: _______

2. Check the outbound processing of the IDoc in the purchase order. What status does the IDoc have? Also note the IDoc number.

   Status: ____

   Number: ________________

3. Display your IDoc in the status monitor as well and send it. What status does the IDoc have now? What is the text for this status?

> Hint:
> You will find the text under the heading *Error text*.

Unit 3
Solution 6

Send a Purchase Order to a Vendor

Business Example

The central purchasing department of the company orders goods from a vendor by means of an EDI connection.

Create a purchase order to an external vendor in the system at company headquarters and check whether an IDoc was generated with the data of this purchase order.

1. In the CORE system, create a purchase order at the vendor T-K515D## for 100 pcs. of the material T-M15A## for the plant 1000. You work for the purchasing organization 1000 and the purchasing group 001 in the company code 1000. Which message type is found, and how is the message to be processed?

   Message type: _______

   Transmission medium: _______

   a) Choose *Logistics* → *Materials Management* → *Purchasing* → *Purchase Order* → *Create* → *Vendor/Supplying Plant Known*.

   b) In the *Vendor* field, enter **`T-K515D##`** and confirm your entry with *Enter*.

   c) On the *Org. Data* tab page, enter the purchasing organization **`1000`**, the purchasing group **`001`**, and the company code **`1000`**.

   d) Choose *Item Overview*, enter the material **`T-M15A##`**, the order quantity **`100`**, the plant **`1000`**, and confirm these entries with *Enter*.

   e) Choose *Messages*. The output type *NEU* was found. The message is to be output by means of *EDI* (transmission medium *6*).

   f) Create the purchase order by saving.

2. Check the outbound processing of the IDoc in the purchase order. What status does the IDoc have? Also note the IDoc number.

   Status: ____

   Number: _______________

   a) Choose *Logistics* → *Materials Management* → *Purchasing* → *Purchase Order* → *Display*.

   b) Choose the *Messages* button. The message should have the status *Successfully processed*. Choose *Back*.

   c) In the menu, choose *System* → *Services for Object* and then *Display relationships*. The number of the outbound IDoc is displayed.

   d) Double-click the number of the IDoc to call it up for display. It has the status *30* (*IDoc is ready for dispatch*).

3. Display your IDoc in the status monitor as well and send it. What status does the IDoc have now? What is the text for this status?

> Hint:
> You will find the text under the heading *Error text*.

a) Choose *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

b) Enter the number from step 2 in the field *IDoc number* and choose *Execute*.

c) Select the row under the output type *ORDERS* and choose the *Process* button.

d) Confirm the message on your selection with *Enter*. The IDoc now has the status *03* (*Data passed to port OK*). The error text is *IDoc written to file*. You can also find the file path using *Error long text*.

Unit 3
Exercise 7

Transfer ALE Scenario Stock Between Distributed Systems (Optional)

Business Example

Sales and Distribution regularly orders stock replenishments at head office for processing its customer orders; head office is where the main warehouse of the company is located. These purchase orders are sent to head office using ALE and are automatically converted into sales orders there.

Create a purchase order with head office in the Sales and Distribution system, check the results of the message determination, and then display the automatically created order in the head office system. Also check the updating of the purchase order after the order confirmation is received.

1. In the Sales system, create a purchase order of the document type NB (*normal order*) at the vendor Core for 100 pcs. of the material DPC1002. You are ordering with the authorization of the purchasing organization 1000 for the plant 1000. You belong to the purchasing group 001. The responsible company code is 1000. Check the results of the message determination: Which message type was found? Which transmission medium is used? Make a note of the document number of the purchase order.

   Message type: ________

   Transmission medium: ______

   Number of the purchase order: _________________

2. Display the outbound IDoc in the status monitor of the Sales sender system. Search for the IDoc using the business object type PurchaseOrder and your document number from step 1. Check the content of individual segments and compare them with your purchase order.

3. In the Core system, check whether a sales order was created for the purchase order of the Sales system. Use your purchase order number from step 1 as the search criterion. Make a note of the document number of the order: ______________

4. Was an order confirmation also created in the Core system and sent to the Sales system? Check the results of the message determination in the order.

5. In the Core system, you now display the outbound IDoc for the order confirmation using the object services in the order. Then check the purchase order in the Sales system: What has changed?

> Hint:
> In the purchase order, check the *Confirmations* tab page on the document item level.

Unit 3
Solution 7

Transfer ALE Scenario Stock Between Distributed Systems (Optional)

Business Example

Sales and Distribution regularly orders stock replenishments at head office for processing its customer orders; head office is where the main warehouse of the company is located. These purchase orders are sent to head office using ALE and are automatically converted into sales orders there.

Create a purchase order with head office in the Sales and Distribution system, check the results of the message determination, and then display the automatically created order in the head office system. Also check the updating of the purchase order after the order confirmation is received.

1. In the Sales system, create a purchase order of the document type NB (*normal order*) at the vendor Core for 100 pcs. of the material DPC1002. You are ordering with the authorization of the purchasing organization 1000 for the plant 1000. You belong to the purchasing group 001. The responsible company code is 1000. Check the results of the message determination: Which message type was found? Which transmission medium is used? Make a note of the document number of the purchase order.

   Message type: ________

   Transmission medium: ______

   Number of the purchase order: _________________

   a) Choose *Logistics* → *Materials Management* → *Purchasing* → *Purchase Order* → *Create* → *Vendor/Supplying Plant Known*.

   b) Enter the vendor **`CORE`**, and on the *Org. Data* tab page, on the document header level, enter the purchasing organization **`1000`**, the purchasing group **`001`** and the company code **`1000`**.

      Hint:
      If the document levels (header or item) are not visible, choose *Header* or *Item Overview* .

   c) On the document item level, enter the material **`DPC1002`**, the order quantity **`100`** and the plant **`1000`**. Confirm your entries with *Enter*.

   d) Choose the *Messages* button. The output type *NEU* was found. The transmission medium is A. Create the purchase order by saving.

2. Display the outbound IDoc in the status monitor of the Sales sender system. Search for the IDoc using the business object type PurchaseOrder and your document number from

step 1. Check the content of individual segments and compare them with your purchase order.

a) Choose *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

b) In the field *Business object*, enter **`PurchaseOrder`**, and in the field *Object key* enter the purchase order number from step 1, then choose *Execute*.

c) An IDoc for the message type *ORDERS* should be found under *IDocs in outbound processing*. Select the entry underneath the message type and choose the *Display IDocs* button.

d) Select the IDoc and choose the *Display IDocs* button again. Open the *Data records* folder to display the segments of the IDoc.

3. In the Core system, check whether a sales order was created for the purchase order of the Sales system. Use your purchase order number from step 1 as the search criterion. Make a note of the document number of the order: _______________

a) Choose *Logistics* → *Sales and Distribution* → *Sales* → *Order* → *Change*.

b) In the *Search Criteria* area, enter your purchase order number from step 1 in the corresponding field and choose *Perform search*.

c) Select the search result and accept it with *Enter*. You now enter the overview screen of the order that was created from the data of the purchase order.

4. Was an order confirmation also created in the Core system and sent to the Sales system? Check the results of the message determination in the order.

a) In the menu of the order, select *Extras* → *Output* → *Header* → *Edit*.

b) The system found the output type *BA00* (*Order Acknowl.*) and immediately processed it into an IDoc when creating the order.

5. In the Core system, you now display the outbound IDoc for the order confirmation using the object services in the order. Then check the purchase order in the Sales system: What has changed?

> Hint:
> In the purchase order, check the *Confirmations* tab page on the document item level.

a) In the sales order, choose *System* → *Services for Object* and then choose *Display relationships*.

b) Double-click the number of the outbound IDoc of the message type *ORDRSP* to call up the IDoc for display.

c) In the system Sales, choose *Logistics* → *Materials Management* → *Purchasing* → *Purchase Order* → *Display*.

d) Select the tab page *Confirmations* on the document item level. In the *Order Ack.* field, you will see the document number of the sales order from the Core system.

Unit 4
Exercise 8

Transfer Trip Cost Accounting with Two Systems

Business Example

Your company has moved the human resources to its own system. Business trips are recorded in the SAP Travel Management of this system and subsequently transferred to the Accounting system for reimbursement purposes.

In the logical system HR_TRAVEL, enter the data of a business trip, settle this business trip, and transfer the settlement data to the logical system CORE for posting in Accounting.

> Caution:
> When the trip costs are being transferred to Accounting, the system checks that the receiver is unique. Therefore, a test run is performed with the model view that the instructor created. Exercise 2, in which you create a new view in the distribution model with all the BAPIs required for the ALE scenario, must therefore be performed after this exercise.

1. In the HR_TRAVEL system, create a new trip for the employee with the personnel number 60995##. Enter the costs for accommodation and taxis. The trip was in the previous week and took two days. Write down the number of the trip: ___________

2. Approve the reimbursement of the trip costs you have just entered and then perform the settlement. Write down the settlement period: ___________

3. Create a posting run with the data for your trip.

4. Now transfer the settlement results to Accounting in the logical system CORE.

5. Then display the inbound IDoc in the status monitor of the logical system Core. You can also directly access your IDoc using the transaction *IDoc Search for Contents* (transaction code `WE09`). Using the IDoc basic type ACC_EMPLOYEE_PAY02, select the segment E1BPACHE06, segment field USERNAME, and enter your user BIT300-## as an additional search criteria.

Transfer Trip Cost Accounting with Two Systems

Business Example

Your company has moved the human resources to its own system. Business trips are recorded in the SAP Travel Management of this system and subsequently transferred to the Accounting system for reimbursement purposes.

In the logical system HR_TRAVEL, enter the data of a business trip, settle this business trip, and transfer the settlement data to the logical system CORE for posting in Accounting.

> Caution:
> When the trip costs are being transferred to Accounting, the system checks that the receiver is unique. Therefore, a test run is performed with the model view that the instructor created. Exercise 2, in which you create a new view in the distribution model with all the BAPIs required for the ALE scenario, must therefore be performed after this exercise.

1. In the HR_TRAVEL system, create a new trip for the employee with the personnel number 60995##. Enter the costs for accommodation and taxis. The trip was in the previous week and took two days. Write down the number of the trip: ___________

   a) Choose *Human Resources* → *Travel Management* → *Travel Expenses* → *Travel Expense Manager*.

   b) Enter the personnel number **60995##** and confirm the entry with *Enter*.

   c) Choose *Create* to call up the screen for entering the trip costs.

   d) In the *Choose a Trip Schema* field, select *Business Trip* and confirm with Enter.

   e) Enter a date in each of the fields *From* and *Finish*. Next, in the *ExpTy* field, enter **HOTL** and in the *Amount* field the total cost of the accommodation. Confirm these entries with *Enter*.

   f) In the *Description* field, enter a text of your choice and select *Enter* again to leave the detail screen.

   g) Next, in the *ExpTy* field, enter **TAXI** and in the *Amount* an amount of your choice. Create the trip by saving and make a note of the trip number.

2. Approve the reimbursement of the trip costs you have just entered and then perform the settlement. Write down the settlement period: ___________

   a) Select your trip from step 1 and choose the *Approve* button.

   b) Then choose the *Settle* button. Confirm the default for the settlement period and make a note of the period.

3. Create a posting run with the data for your trip.
   a) Choose *Human Resources → Travel Management → Travel Expenses → Periodic Processing → Transfer to Accounting → Create Posting Run*.
   b) Enter the payroll area **D2**, choose *Other period* and enter the period from step 2. Also enter the personnel number **60995##** and the number of the trip from step 1 and choose *Execute*.
   c) If the synchronous check of the accounts, account assignment objects and HR payee assignment for the personnel number was successful, you will see a green traffic light for the number of the posting run. Leave the transaction.
4. Now transfer the settlement results to Accounting in the logical system CORE.
   a) Choose *Human Resources → Travel Management → Travel Expenses → Periodic Processing → Transfer to Accounting → Manage Posting Runs*.
   b) Select *Execute* to display your posting run from step 3.
   c) Select your posting run and choose the *Post* button. Answer the query on the processing time by choosing the *Post Immed.* button: If the transfer was successful, you will see a green traffic light with the text *Transfer Complete Posting Run to target system CORE*.
5. Then display the inbound IDoc in the status monitor of the logical system Core. You can also directly access your IDoc using the transaction *IDoc Search for Contents* (transaction code `WE09`). Using the IDoc basic type ACC_EMPLOYEE_PAY02, select the segment E1BPACHE06, segment field USERNAME, and enter your user BIT300-## as an additional search criteria.
   a) In the Core system, choose *Tools → ALE → ALE Administration → Monitoring → IDoc Display → Status Monitor* to call the status monitor, or *Tools → ALE → ALE Administration → Services → IDoc Search for Contents* to go to transaction `WE09`.
   b) In the field *Basic type* enter **ACC_EMPLOYEE_PAY02**, in the field *Search in Segment* enter **E1BPACHE06**, in the field *Search in Field* enter **USERNAME** and in the field *For Value* enter your user **BIT300-##** and choose *Execute*.
   c) Click on the number of the IDoc to display the details.

Unit 4
Exercise 9

Set Up Travel Expenses with ALE Scenario

Business Example

Your company has moved the human resources to its own system. Business trips are recorded in the SAP Travel Management of this system and subsequently transferred to the Accounting system for reimbursement purposes.

Task 1:

In the IMG documentation, search for the description of the ALE scenario Transferring the accounting results of the travel expenses to Accounting and make a note of the BAPIs required for this scenario.

1. In the IMG documentation, search for the description of the ALE scenario Transferring the accounting results of the travel expenses to Accounting.

2. Which BAPIs are involved in the transfer of the trip costs to Accounting? Make a note of the BAPIs for the business object types used for the steps Create posting run, Check posting run, and Post posting run.

   ____

   ____

   ____

3. In the BAPI Explorer, check for which of the BAPIs entered in the above table there is an ALE interface. Note the names of the assigned message types.

BAPI	Message type

Task 2:

In the logical system Core, define a new logical system Core## and then create a model view with this logical system as the target system of the method calls determined in the first task.

> Hint:
> To be able to perform this exercise with different exercise groups, we have to use a little trick: All the groups are to set up the distribution between the systems HR_TRAVEL and Core. However, just like message types, BAPIs may only appear in one view of the distribution model in the relationship between two logical systems. For this reason, you use the logical system Core## as the target system, but you enter a port to which an RFC destination of the system Core is assigned.

1. In the Core logical system , you define the logical system CORE##, but you do not assign it to any client.

2. Is this logical system name now known in the logical system HR_TRAVEL? If applicable, also create CORE## in HR_TRAVEL.

3. In the distribution model of the system Core, create a new view with the short text BIT300BAPI## and the technical name BAPI##. Add the BAPIs that you determined in the first task. The sender is the system HR_TRAVEL, and the receiver the system CORE##.

Task 3:

Distribute the new model view to the system HR_TRAVEL and there you create the corresponding partner profiles. You also add an RFC destination to the system CORE## for synchronous method calls.

1. Distribute the new model view to the logical system HR_TRAVEL and check the result.

> Caution:
> Do not distribute the view to the logical system CORE##, because this logical system is not assigned to a client.

2. In the logical system HR_TRAVEL, create outbound partner profiles with the partner CORE## for the message type SYNCH. The receiving port is Core with the RFC destination of the same name.

3. In the logical system HR_TRAVEL, you now generate the outbound partner profiles with the system CORE## for the three asynchronous method calls, or you create them manually.

4. In the logical system Core, check the inbound partner profiles with the logical system HR_TRAVEL for the message types for asynchronous communication. Which process code is assigned? Why do you not need any separate inbound partner profiles for the logical system CORE##?

5. In the HR_TRAVEL system, which RFC destination do you have to assign to the logical system CORE## for synchronous method calls? Add the corresponding entry.

Set Up Travel Expenses with ALE Scenario

Business Example

Your company has moved the human resources to its own system. Business trips are recorded in the SAP Travel Management of this system and subsequently transferred to the Accounting system for reimbursement purposes.

Task 1:
In the IMG documentation, search for the description of the ALE scenario Transferring the accounting results of the travel expenses to Accounting and make a note of the BAPIs required for this scenario.

1. In the IMG documentation, search for the description of the ALE scenario Transferring the accounting results of the travel expenses to Accounting.
   a) Call the transaction `SALE` and choose *IDoc Interface/Application Link Enabling (ALE) → Modelling and Implementing Business Processes → Configure Predefined ALE Business Processes → Human Resources → HR → AC → Transfer Travel Expenses to FI → Travel Management and Accounting are Release 4.5A or Later*.

   b) Choose *IMG activity documentation*.

2. Which BAPIs are involved in the transfer of the trip costs to Accounting? Make a note of the BAPIs for the business object types used for the steps Create posting run, Check posting run, and Post posting run.

---

3. In the BAPI Explorer, check for which of the BAPIs entered in the above table there is an ALE interface. Note the names of the assigned message types.

BAPI	Message type
AcctngEmplyeeExpnses.Post	*ACC_EMPLOYEE_EXP*
AcctngEmplyeePaybles.Post	*ACC_EMPLOYEE_PAY*
AcctngEmplyeeRcvbles.Post	*ACC_EMPLOYEE_REC*

   a) Choose *Tools → Business Framework → BAPI Explorer*.

   b) Select the tab page *Alphabetical* and expand each of the entries for the business object types AcctngEmplyeeExpnses, AcctngEmplyeePaybles and AcctngEmplyeeRcvbles.

   c) Select each respective entry for the method Post. You will find the message type on the tab page *Detail* in the field *ALE message type*.

Task 2:
In the logical system Core, define a new logical system Core## and then create a model view with this logical system as the target system of the method calls determined in the first task.

> Hint:
> To be able to perform this exercise with different exercise groups, we have to use a little trick: All the groups are to set up the distribution between the systems HR_TRAVEL and Core. However, just like message types, BAPIs may only appear in one view of the distribution model in the relationship between two logical systems. For this reason, you use the logical system Core## as the target system, but you enter a port to which an RFC destination of the system Core is assigned.

1. In the Core logical system , you define the logical system CORE##, but you do not assign it to any client.
   a) Call the transaction `SALE` and choose *IDoc Interface / Application Link Enabling (ALE)* → *Basic Settings* → *Logical Systems* → *Define Logical System*. Confirm the message about client independence with *Enter*.
   b) Choose the *New Entries* button, enter the logical system name **`CORE##`** and a name of your choice and save your entry.
   c) In the screen prompting the workbench request, choose *Create Request*, enter a short description of your choice and choose *Save*. Then choose *Continue* or *Enter* and leave the table.
2. Is this logical system name now known in the logical system HR_TRAVEL? If applicable, also create CORE## in HR_TRAVEL.

   This depends on whether you are working with one or more SAP systems in the course.
3. In the distribution model of the system Core, create a new view with the short text BIT300BAPI## and the technical name BAPI##. Add the BAPIs that you determined in the first task. The sender is the system HR_TRAVEL, and the receiver the system CORE##.
   a) Call the transaction `SALE` and choose *IDoc Interface / Application Link Enabling (ALE)* → *Modelling and Implementing Business Processes* → *Maintain Distribution Model and Distribute Views*.
   b) Switch to the change mode (*Switch between display and edit modes*) and choose the *Create model view* button.
   c) Enter the short text **`BIT300BAPI##`** and the technical name **`BAPI##`** and choose *Enter*.
   d) Select the view and choose the *Add BAPI* button.
   e) Enter the sender **`HR_TRAVEL`**, the receiver **`CORE##`**, the object name **`AcctngEmplyeeExpnses`** and the method **`Check`**, and confirm your entries with *Enter*.
   f) In this way, you add the remaining BAPIs that you require for the ALE scenario. Create the new model view by saving.

Task 3:
Distribute the new model view to the system HR_TRAVEL and there you create the corresponding partner profiles. You also add an RFC destination to the system CORE## for synchronous method calls.

1. Distribute the new model view to the logical system HR_TRAVEL and check the result.

> Caution:
> Do not distribute the view to the logical system CORE##, because this logical system is not assigned to a client.

   a) In the distribution model of the system Core, select the new model view and choose *Edit → Model view → Distribute*.

   b) In the next screen, deselect the entry for the logical system *CORE##* and confirm the change with *Enter*. You receive a log of the successful distribution of the view to the system HR_TRAVEL.

   c) In the system HR_TRAVEL, call the distribution model for display, as described in task 2.

> Hint:
> If your BAPI## view has not been distributed, this could be because the message type SYNCH does not exist in the partner profiles of the system CORE with the system HR_TRAVEL, or because the RFC destination does not lead to the client of the system HR_TRAVEL.

2. In the logical system HR_TRAVEL, create outbound partner profiles with the partner CORE## for the message type SYNCH. The receiving port is Core with the RFC destination of the same name.

   a) In the system HR_TRAVEL, in the menu of the distribution model, choose *Environment → Change Partner Profile*.

   b) Choose *Create*, enter the partner number **CORE##** and the partner type **LS** and create the new partner by saving.

   c) In the outbound parameter area, choose *Create outbound parameter*, and the message type **SYNCH**, enter the receiver port **CORE** and the IDoc type **SYNCHRON**. Choose the immediate IDoc transfer and save your changes. Leave the partner profile maintenance.

3. In the logical system HR_TRAVEL, you now generate the outbound partner profiles with the system CORE## for the three asynchronous method calls, or you create them manually.

   a) Select your model view and in the menu of the distribution model, choose *Environment → Generate Partner Profile*.

   b) Choose *Environment → Change Partner Profile* to check the automatically created partner profiles.

   c) Create the partner profiles manually, choose *Environment → Change Partner Profile*.

   d) Select the partner *CORE##* and in the outbound parameter area, choose *Create outbound parameter*.

e) Enter the message type **ACC_EMPLOYEE_EXP**, the port **CORE** and the IDoc type **ACC_EMPLOYEE_EXP02**, select the immediate processing, and create the partner profiles by saving. Proceed in the same way with the message types ACC_EMPLOYEE_REC and ACC_EMPLOYEE_PAY. The corresponding IDoc types are ACC_EMPLOYEE_REC02 and ACC_EMPLOYEE_PAY02 respectively.

4. In the logical system Core, check the inbound partner profiles with the logical system HR_TRAVEL for the message types for asynchronous communication. Which process code is assigned? Why do you not need any separate inbound partner profiles for the logical system CORE##?

   In the inbound partner profiles for the message types ACC_EMPLOYEE_EXP, ACC_EMPLOYEE_PAY and ACC_EMPLOYEE_REC, the process code BAPI is assigned. Because the logical system CORE## is not assigned to any client, but rather the system HR_TRAVEL actually calls the system CORE## on the basis of the assignment of the receiving port CORE in the outbound partner profiles with CORE##, no inbound partner profiles are required for CORE##.

5. In the HR_TRAVEL system, which RFC destination do you have to assign to the logical system CORE## for synchronous method calls? Add the corresponding entry.

   a) Because the logical system name CORE## is not assigned to any client, you must assign the RFC destination Core to the logical system of the same name.

   b) Call the transaction `SALE` and choose *IDoc Interface/Application Link Enabling (ALE)* → *Communication* → *Determine RFC Destinations for Method Calls*.

   c) In the menu of the transaction, choose *Edit* → *Find*. In the *Search for* field, enter your logical system **CORE##** and choose *Find* or *Enter*.

   d) Click on the system name to return to the system overview at the position of the corresponding entry.

   e) Select your logical system CORE## and choose the button *Standard BAPI destination*. Enter the RFC destination **CORE**, choose *Enter*, and save the change.

Unit 5
Exercise 10

Trace IDocs in the Status Monitor

Business Example

You have set up various ALE scenarios in your system landscape. Now you want to check whether the IDocs sent for test purposes have arrived in the respective receiving systems without logging on to these receiving systems.

In the logical system Core, display IDocs for the message type *MATMAS* and check whether they have been processed successfully in the logical system Production. However, first check the assignment of an RFC destination for dialogs to the logical system Production.

1. In the Core system, check whether the Production system is assigned an RFC destination for dialogs.
2. In the Core system, call the status monitor and select the IDocs of the message type *MATMAS* that have been created since the start of the course. Use the function, TraceIDocs, to determine the status of these IDocs in the receiving system Production. Display one of the IDocs in the Production system.

Unit 5
Solution 10

Trace IDocs in the Status Monitor

Business Example

You have set up various ALE scenarios in your system landscape. Now you want to check whether the IDocs sent for test purposes have arrived in the respective receiving systems without logging on to these receiving systems.

In the logical system Core, display IDocs for the message type *MATMAS* and check whether they have been processed successfully in the logical system Production. However, first check the assignment of an RFC destination for dialogs to the logical system Production.

1. In the Core system, check whether the Production system is assigned an RFC destination for dialogs.
   a) Call the transaction `SALE` and choose *IDoc Interface / Application Link Enabling (ALE)* → *Communication* → *Determine RFC Destinations for Method Calls*.
   b) In the menu of the transaction, choose *Edit* → *Find*. In the *Find* field, enter the logical system **`PRODUCTION`** and choose *Find* or *Enter*.
   c) Click on the system name to return to the system overview at the position of the corresponding entry. The logical system Production is assigned the RFC destination for dialog calls of the same name.
2. In the Core system, call the status monitor and select the IDocs of the message type *MATMAS* that have been created since the start of the course. Use the function, TraceIDocs, to determine the status of these IDocs in the receiving system Production. Display one of the IDocs in the Production system.
   a) Choose *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.
   b) Enter the message type **`MATMAS`**, the partner system **`PRODUCTION`**, and as the change date, the starting date of the course, and choose *Execute*.
   c) Select the entry (or one of the entries) underneath the message type and choose *Trace Idocs*.
   d) Check the status of the IDoc(s). If the inbound processing was successful, you will find the status *53* beside the IDoc number.
   e) Double-click on the number of the inbound IDoc in the Production system. You enter the display of the IDoc in the receiving system.

Unit 5
Exercise 11

Carry Out an ALE Audit

Business Example

You want to be able to detect by the status value of an ALE scenario in the sending system whether the transfer of the data to the corresponding application in the receiving system was successful, or whether errors occurred.

Test the ALE Audit using your message type ZMATMAS##.

1. Send an IDoc with the status confirmations from the system Production to the system Core. Only send confirmations for IDocs for your reduced message type ZMATMAS##, namely, for all those created since the start of the course.

2. Check the result of the confirmation in the status monitor of the Core system and display one of the IDocs involved.

Carry Out an ALE Audit

Business Example

You want to be able to detect by the status value of an ALE scenario in the sending system whether the transfer of the data to the corresponding application in the receiving system was successful, or whether errors occurred.

Test the ALE Audit using your message type ZMATMAS##.

1. Send an IDoc with the status confirmations from the system Production to the system Core. Only send confirmations for IDocs for your reduced message type ZMATMAS##, namely, for all those created since the start of the course.

   a) In the Production system, choose *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

   b) Choose *Execute* .

   c) In the menu of the status monitor, choose *Goto* → *ALE Audit* → *Send Audit Confirmations*.

   d) In the field *Confirmations to system*, enter **CORE**, in the field *Message type*, enter **ZMATMAS##**, and in the fields *Date IDoc changed*, enter the date on which the course started and today's date. Then choose *Execute* .

   e) Confirm the message on the creation of the audit IDoc, return to the initial screen of the status monitor and update the display using *Refresh IDoc display* . You see an entry in the area of the outbound IDoc for the message type *ALEAUD*.

2. Check the result of the confirmation in the status monitor of the Core system and display one of the IDocs involved.

   a) In the Core system, call the status monitor as described in step 1a), enter the message type **ZMATMAS##** and choose *Execute* . Your outbound IDocs for the message type *ZMATMAS##* now have different status values than before. If the processing was successful in the application in the Production system, you will see the status *41*, otherwise you see the status *39*, which may also mean that your IDoc has not been transferred to the application yet. If the outbound processing contained errors, the status is set to *40*.

   b) Select the entry underneath the message type and choose *Display IDocs*.

   c) Select an IDoc and choose *Display IDocs*. Open the folder of the status records in order to monitor the updating of the status values.

Unit 5
Exercise 12

Analyze and Handle Errors

Business Example

In the logical system Core, you have created or changed master data and distributed it to the receiving systems Production and Sales. The data transfer is not always successful. Check the situation in the status monitor and resolve the errors.

Task 1: Analysis of the IDoc Status Values

In the case of incorrect IDoc processing, analyze the status values of the IDoc involved to determine in which part of the process an error occurred.

1. Write down the status values belong to each (error) situation.

   The IDoc was not created.

   ____

   ____

   ____

   The IDoc was created, but was not sent.

   ____

   ____

   ____

   The IDoc was successfully sent, but it was not received by the receiving system.

   ____

   ____

   ____

   The IDoc was received by the receiving system, but was not processed yet.

   ____

   ____

   ____

Task 2: Postprocessing of an Incorrect IDoc

The IDoc inbound processing can fail because of discrepancies between the application-specific Customizing settings in the sending and receiving systems. The IDoc is given the status 51 (*Application document not posted*). In this exercise you will learn how to postprocess an incorrect IDoc so that its data can be posted in the application.

1. In the logical system, Core, add the division 99 to your material master, MATALE-##.

2. Send the changed material to the logical system Production and display the outbound IDoc in the status monitor. What status does the outbound IDoc have? What status does the inbound IDoc have in the receiving system?

> Hint:
> Use the function *Monitor IDocs* in the Core system to gain information about the inbound processing in the Production system.

3. In the Production system, call up and display the IDoc in the status monitor and find out the cause of the processing error. Make a note of the number of the IDoc: __________

4. What conclusion do you draw from the error message? What options do you have for solving the problem?

____

____

____

5. Call the incorrect IDoc for display, switch to the change mode, and replace the value 99 in the field *Division* with a value defined in the Production system.

> Hint:
> In doing this, use the input help for the field *Division*.

6. Now try to transfer the IDoc to the application again. What happens? Check the status records of your IDoc.

Task 3: Setting a Deletion Flag for an IDoc

In some cases, it is no longer possible, or is not desirable, to handle an error by correcting system settings or editing the IDoc. You want the IDoc to be archived unchanged. In such cases, you can flag the IDoc for deletion so that it can be archived at the next opportunity.

1. Send your changed material *MATALE-##* from the Core system to the Production system again and call up the status monitor there.

2. In the Production system, your IDoc has the status 51 again. Instead of editing it, you now set a deletion flag for the subsequent archiving.

Unit 5
Solution 12

Analyze and Handle Errors

Business Example

In the logical system Core, you have created or changed master data and distributed it to the receiving systems Production and Sales. The data transfer is not always successful. Check the situation in the status monitor and resolve the errors.

Task 1: Analysis of the IDoc Status Values

In the case of incorrect IDoc processing, analyze the status values of the IDoc involved to determine in which part of the process an error occurred.

1. Write down the status values belong to each (error) situation.

   The IDoc was not created.

   No status value: An IDoc only gets a status once it has been saved on the database.

   The IDoc was created, but was not sent.

   Both status 30 (*IDoc is ready for dispatch)* and status 02 (*Error passing data to port*) could apply here.

   The IDoc was successfully sent, but it was not received by the receiving system.

   The IDoc has the status 03 (*data passed to port OK*) and also, if the program RBDMOIND was executed after it was sent, the status 12 (*Dispatch OK*).

   The IDoc was received by the receiving system, but was not processed yet.

   If the ALE Audit is set up, the IDoc in the sending system has the status 39 (*IDoc is in the target system*). In the receiving system the IDoc has the status 64 (*IDoc ready to be transferred to application*).

Task 2: Postprocessing of an Incorrect IDoc

The IDoc inbound processing can fail because of discrepancies between the application-specific Customizing settings in the sending and receiving systems. The IDoc is given the status 51 (*Application document not posted*). In this exercise you will learn how to postprocess an incorrect IDoc so that its data can be posted in the application.

1. In the logical system, Core, add the division 99 to your material master, MATALE-##.

a) Choose *Logistics* → *Materials Management* → *Material Master* → *Material* → *Change* → *Immediately*.

b) Enter the material number **MATALE-##** and choose *Enter*.

c) Select the view *Basic data 1* and choose *Enter*.

d) In the field *Division*, enter **99** and save the change.

2. Send the changed material to the logical system Production and display the outbound IDoc in the status monitor. What status does the outbound IDoc have? What status does the inbound IDoc have in the receiving system?

> Hint:
> Use the function *Monitor IDocs* in the Core system to gain information about the inbound processing in the Production system.

a) Choose *Tools* → *ALE* → *Master Data Distribution* → *Cross-Application* → *Material* → *Send*.

b) Enter your material **MATALE-##**, your reduced message type **ZMATMAS##**, and the logical system **PRODUCTION** and choose *Execute*.

c) Choose *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

d) Enter your message type **ZMATMAS##** and choose *Execute* . The IDoc has the status 03.

e) Select the entry underneath the message type and choose *Trace IDocs*. In the system Production, the IDoc has the status 51.

3. In the Production system, call up and display the IDoc in the status monitor and find out the cause of the processing error. Make a note of the number of the IDoc: __________

a) Choose *Tools* → *ALE* → *ALE Administration* → *Monitoring* → *IDoc Display* → *Status Monitor*.

b) Enter your message type **ZMATMAS##** and choose *Execute* .

c) Select the entry underneath the message type and choose *Display IDocs*.

d) Select the IDoc and choose *Display status long text*. You will see that an application log has been written from which you can obtain more detailed information about the problem.

e) Copy the number of the log and click on *Proceed*.

f) In the field *Ext. number of application log*, enter this number and choose *Execute*.

g) Double-click on the entry with the red traffic light (*Problem Class: Very Important)*. You will see that the value 99 is not permitted for the field *MARA-SPART* (*division*). Leave the log and the display of the error long text.

4. What conclusion do you draw from the error message? What options do you have for solving the problem?

   Division 99, which you entered in the material master in the sending system before the distribution, is not defined in the receiving system. Therefore, the Customizing settings of the two systems are inconsistent. Now you can either also define division 99 in the receiving system and have the IDoc processed again, or you can edit the IDoc and have a value defined in the receiving system for the field *Division* transferred to the application. In the exercise, you edit your IDoc.

5. Call the incorrect IDoc for display, switch to the change mode, and replace the value 99 in the field *Division* with a value defined in the Production system.

   > Hint:
   > In doing this, use the input help for the field *Division*.

   a) Choose *Display IDocs*.

   b) Open the *Data records* tab page and double-click on the symbol to the left of the segment *E1MARAM*.

   c) Choose *Data Record* → *Display* → *Change* and confirm the information that your changes have been written to the database with *Enter*.

   d) Navigate to the field *SPART (division)* and call up the input help. Use a double-click to accept one of the entries in the list of permitted values. Save this change, leave the data record editor, and then leave the IDoc display.

6. Now try to transfer the IDoc to the application again. What happens? Check the status records of your IDoc.

   a) Go back to the status monitor and choose *Refresh IDoc display*. The system sets the status 69 (*IDoc was edited*).

   b) Select the entry for your edited IDoc and choose *Process*. The status of the IDoc has been changed from 69 to 53.

   c) Choose *Display IDoc* again and open the folder *Status records*. You see that after the editing, your IDoc has been reset to status 51 and then transferred to the application again.

   d) Go from the IDoc display back to the status monitor. The system has created a second IDoc and assigned to it the status 70 (*Original of an IDoc which was edited*). The long text is, *This IDoc is saved as the original of an edited document.* Thus, the system keeps a copy of the incorrect IDoc as it was before the editing.

   e) Call up the edited IDoc for display again, open the folder of the status records and double-click on the symbol to the left of the status long text for the status 69. You will see which user edited this IDoc in which segment, and which number the copy with the original data has.

   f) Leave the IDoc display, select your IDoc and choose *Display relationships*. A screen appears with the two IDoc numbers.

Task 3: Setting a Deletion Flag for an IDoc

In some cases, it is no longer possible, or is not desirable, to handle an error by correcting system settings or editing the IDoc. You want the IDoc to be archived unchanged. In such cases, you can flag the IDoc for deletion so that it can be archived at the next opportunity.

1. Send your changed material *MATALE-##* from the Core system to the Production system again and call up the status monitor there.
    a) Proceed in the way described in the second task in step 2, in order to distribute your material *MATALE-##*.
    b) Find the incorrect IDoc in the Production system in the status monitor (see step 3). It has the status 51 again.
2. In the Production system, your IDoc has the status 51 again. Instead of editing it, you now set a deletion flag for the subsequent archiving.
    a) Select your IDoc in the status monitor and choose *Edit* → *Restrict and process*.
    b) Remove the indicator *Background processing* and choose *Execute* .
    c) Choose *Delete flag* and confirm the confirmation prompt with *Yes*: A message appears that your IDoc has been flagged for deletion.
    d) Check the result in the status monitor. The IDoc now has the status 68 (*Error, no further processing*).

Unit 5
Exercise 13

Process an Incorrect IDoc from a Work Item

Business Example

You have configured ALE scenarios in your system landscape. If errors occur, an administrator or an employee of the business department should be quickly informed by means of a workflow.

Transfer the message type ZCREMAS## into your model view TRAINING##, generate the outbound partner profiles with the partner system PRODUCTION and distribute the model view again. Then send a vendor master and check the inbound IDoc. Find the corresponding work item and remove the error by means of the work item processing.

1. In the logical system CORE, in your model view TRAINING##, add an entry for the message type ZCREMAS##. The sender is the logical system CORE, and the receiver is PRODUCTION. Generate the outbound partner profiles for this message type and distribute your model view again to the system PRODUCTION.

2. Send the master record of the vendor T-K515D## from the system Core to the system Production. Check the inbound IDoc in the status monitor of the receiving system. What status does it have? Make a note of the number of the IDoc: ________________

> Hint:
> Use the message type **ZCREMAS##**.

3. Check in the system Production whether your inbox in the SAP Business Workplace contains a work item for this IDoc. Display the IDoc from the attachment of the work item, determine the cause of the error, and add the missing settings.

4. Try to restart the processing of the IDoc. What status does the IDoc have now?

Process an Incorrect IDoc from a Work Item

Business Example

You have configured ALE scenarios in your system landscape. If errors occur, an administrator or an employee of the business department should be quickly informed by means of a workflow.

Transfer the message type ZCREMAS## into your model view TRAINING##, generate the outbound partner profiles with the partner system PRODUCTION and distribute the model view again. Then send a vendor master and check the inbound IDoc. Find the corresponding work item and remove the error by means of the work item processing.

1. In the logical system CORE, in your model view TRAINING##, add an entry for the message type ZCREMAS##. The sender is the logical system CORE, and the receiver is PRODUCTION. Generate the outbound partner profiles for this message type and distribute your model view again to the system PRODUCTION.

   a) Call the transaction `SALE` and choose *IDoc Interface / Application Link Enabling (ALE) → Modelling and Implementing Business Processes → Maintain Distribution Model and Distribute Views*.

   b) Switch to the change mode (*Switch between display and change mode*) and in your model view TRAINING## select the entry for the receiving system PRODUCTION and choose the button *Add message type*.

   c) Enter the message type **ZCREMAS##** and choose *Enter*. Save your changes.

   d) Select your model view TRAINING##, and in the menu of the distribution model, choose *Environment → Generate Partner Profiles* and in the next screen, choose *Execute*. Outbound parameters are created for the message type ZCREMAS##. Leave the log display.

   e) Select your model view *TRAINING##* and in the menu of the distribution model, choose *Edit → Model View → Distribute*. Deselect the entry for the logical system SALES and choose *Enter*.

2. Send the master record of the vendor T-K515D## from the system Core to the system Production. Check the inbound IDoc in the status monitor of the receiving system. What status does it have? Make a note of the number of the IDoc: ______________

   > Hint:
   > Use the message type **ZCREMAS##**.

   a) Choose *Tools → ALE → Master Data Distribution → Cross-Application → Vendor → Send*.

b) In the field *Account number of the vendor*, enter **T-K515D##**, in the field *Message type*, enter **ZCREMAS##** and in the field *Target system*, enter **PRODUCTION** and choose *Execute*. A master IDoc and a communication IDoc are created.

c) In the Production system, choose *Tools → ALE → ALE Administration → Monitoring → IDoc Display → Status Monitor*.

d) Enter your message type **ZCREMAS##** and choose *Execute*: Your IDoc has the status 03 (*IDoc with errors added*).

e) Select the entry and click on the *Display IDocs* button. Make a note of the number of the IDoc.

3. Check in the system Production whether your inbox in the SAP Business Workplace contains a work item for this IDoc. Display the IDoc from the attachment of the work item, determine the cause of the error, and add the missing settings.

a) Choose *Menu → Business Workplace*.

b) Choose *Inbox → Workflow → Grouped according to content*. Search the list for your IDoc from step 2 e) and select it so that your work item appears in the detail window on the right half of the screen.

c) Choose *Execute*, and in the next screen (the status display of the IDoc) choose the button *IDoc display*.

d) Open the folder *Status Records* and double-click on the symbol beside the status long text (*EDI: Partner profile inbound not available*).

e) The error analysis tells you that the inbound partner profiles with the partner PRODUCTION are missing for the message type ZCREMAS##. Click on the text *Execute function*.

f) The system takes you to the transaction for maintaining partner profiles. Select the system CORE in the folder of the partner type LS, and in the inbound parameter area, choose *Create inbound parameter*.

g) Enter the message type **ZCREMAS##** and the process code **CRE1** and save the change. Leave the maintenance of the partner profiles, close the display of the status long text and leave the IDoc display.

4. Try to restart the processing of the IDoc. What status does the IDoc have now?

a) In the status record display, choose the button *Process*. A message tells you that your IDoc was processed successfully.

b) Confirm the message with *Enter* and choose *Refresh*: The work item has been deleted from your inbox.

c) Call up the status monitor again. You will find a description of the procedure in step 2 c). If your status monitor was still open in another mode, choose *Refresh IDoc display*: The IDoc now has the status 53 (*Application document posted*).